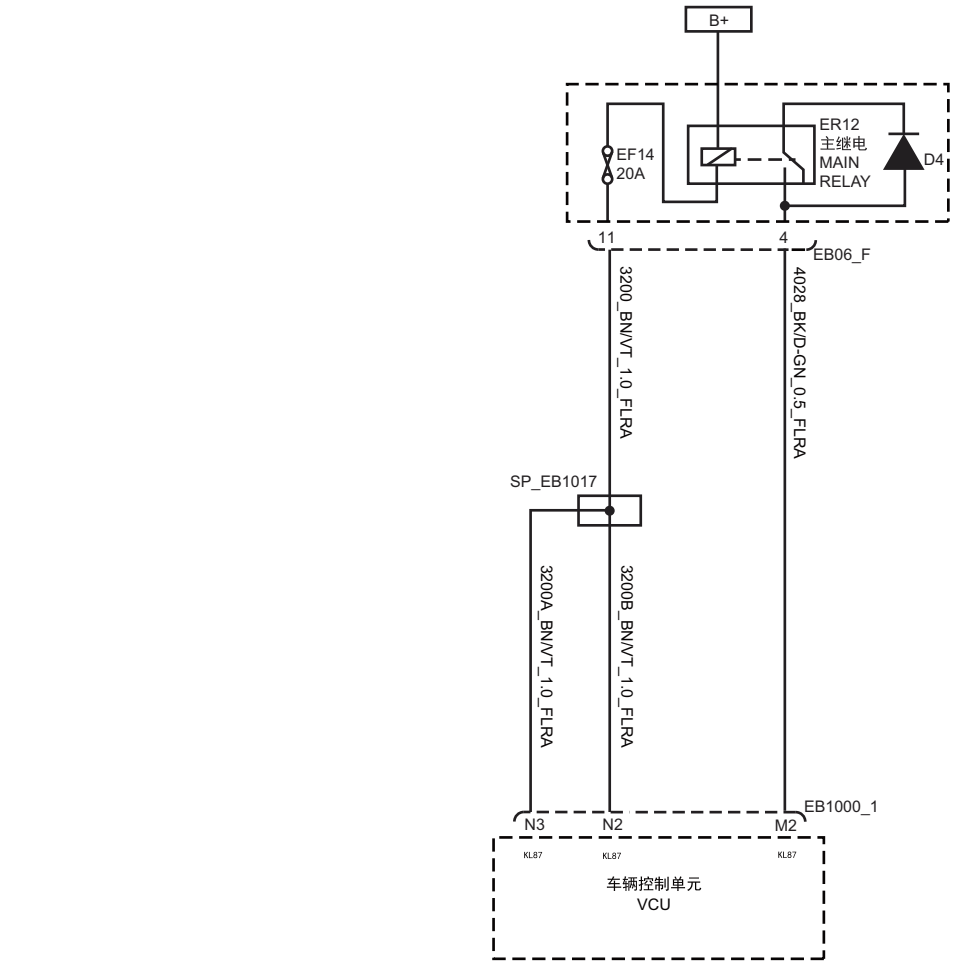
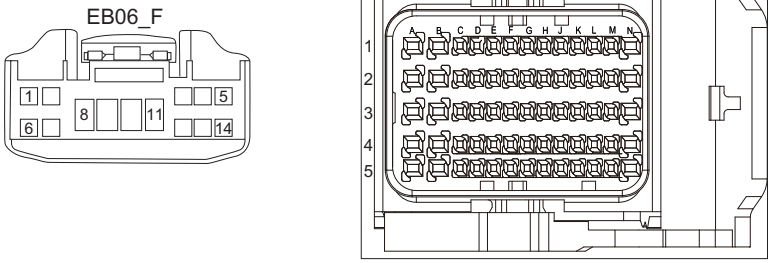
P0687、 P0686、 P0688

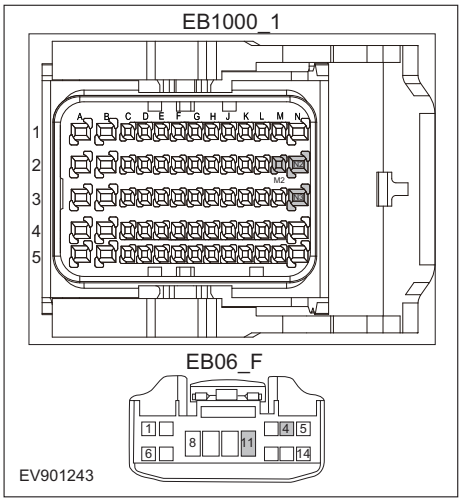
Fault diagnosis code
P0687：DEM_stMainRelaySBCfmd
P0686：DEM_stMainRelaySGCfmd
P0688：DEM_stMainRelayOCCfmd
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • Main relay fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0687
The underlying diagnosis
P0686
The underlying diagnosis
P0688
The underlying diagnosis
To set the effect of a fault code condition
Description of circuit diagram

Circuit diagram





EV901209

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse EF14 for damage. 2 Check the relay ER12 for damage. 3 Check battery capacity. 4 Check the harness plug of main relay, VCU for damage, poor contact, aging and looseness. 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the VCU with diagnostic apparatus. 4 Is there any other fault code except for P0687/P0686/P0688?
		YES → Refer to:DTC Summary list(VCU).
NO ↓		
Step 3	Check the circuit between main relay and VCU (open circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB06_F of UEC. 3 Disconnect the harness plug EB1000_1 of GAW module. 4 Measure the resistance between the harness terminal of UEC and the VCU. Measuring circuit: resistance between terminal 4 on plug EB06_F and terminal M2 on plug EB1000_1. Measuring circuit: resistance between terminal 11 on plug EB06_F and terminal N2 on plug EB1000_1. Measuring circuit: resistance between terminal 11 on plug EB06_F and terminal N3 on plug EB1000_1. **Standard value: ≈ 0Ω** 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the main relay. (components)
	1 Turn the ignition switch to OFF. 2 Replace the main relay in good condition. 3 Connect the fault diagnosis instrument to the diagnosis interface. 4 Turn the ignition switch to ON position. 5 Clear fault codes. 6 Reread fault codes. 7 Test whether the faulted code is existed.
	NO → Replace the main relay.
YES ↓	
Step 5	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	